321311

12425

3 Hours / 70 Marks Seat No. | | | | | | | | |

Instructions : (1) All Questions are *compulsory.*

(2) Answer each next main Question on a new page.

(3) Illustrate your answers with neat sketches wherever necessary.

(4) Figures to the right indicate full marks.

(5) Mobile Phone, Pager and any other Electronic Communication devices are not permissible in Examination Hall.

Marks

1. Attempt any FIVE of the following : 10

(a) Define Enzymes.

(b) Define organic compounds.

(c) Enlist types of Nucleic acid with full form.

(d) Draw homologous series of 10-Alkanes.

(e) Define molecular formula.

(f) Define peptide bond.

(g) Enlist the types of lipids with examples.

2. Attempt any THREE of the following : 12

(a) Write down properties and characteristics of various radioactive compounds.

(b) Explain the significance of acid, saponification and Richert Missel values.

(c) Explain in detail isomerism with examples.

(d) Write down the laboratory safety rules.

3. Attempt any THREE of the following : 12

(a) Differentiate between alpha, beta and gamma rays.

(b) Define Biomedical waste and explain the disposal rules and regulation.

(c) Identify types of chemicals based on symbol.

(d) Write rules of IUPAC nomenclature of aliphatic compounds.

4. Attempt any THREE of the following : 12

(a) Write down introduction and uses of carbohydrates.

(b) Write down introduction, definition and uses of enzymes.

(c) Describe various types of glassware used in medical laboratory.

(d) Explain terms Isotope, Isobar and Isotone with examples.

(e) Describe Nuclear Fusion and Fission reaction.

5. Attempt any TWO of the following : 12

(a) Define amino acid and classify it.

(b) Write about different types of Yantra used by Nagarjuna Swedani, Patana, Adhapatan, Kosthi and Dola Yantra.

(c) Describe types and characteristics of hybridization.

6. Attempt any TWO of the following : 12

(a) Write down method of preparation from alkyl halide-oxygen-alcohol.

(b) Explain types of distillation-simple and fractional.

(c) Write down about radioactivity measurement and uses of radioactivity in medical field.